

Simultaneous Inferences — Restaurant Tips

From one interval to a family of statements

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<https://github.com/HelloJJJie/Assignment-Forecast-and-Analysis-of-Regression.git>

A dataset of $n = 244$ restaurant bills — how sure can we be about *several* statements at the same time?

1. Why simultaneous inference? Joint intervals for β_0, β_1

Data. tips.csv, X = total bill, Y = tip (USD);
 $\hat{Y} = 0.920 + 0.105X$.

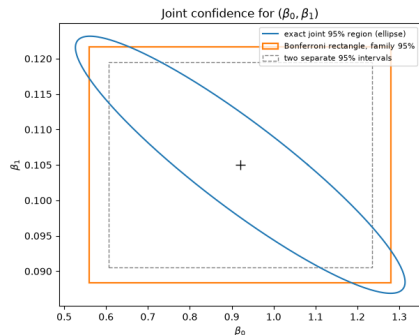
Problem. A 95% interval is right *one statement at a time*. For g statements, $P(\text{all correct}) \geq 1 - g\alpha$ (Bonferroni), so each interval needs level $1 - \alpha/g$.

Share of runs in which the interval(s) cover the truth:

Intervals	mult.	β_0	β_1	both
two separate 95%	1.970	.945	.947	.927
Bonferroni, fam. 95%	2.255	.973	.975	.962

Simulated: 5000 synthetic data sets drawn from the fitted model (normal errors, same X); Monte-Carlo s.e. ≈ 0.003 .

Separate intervals fall *below* 95% jointly; Bonferroni guarantees at least 95% and here lands at 96.2% — conservative, but safe.



	single 95%	family 95%
β_0	[0.606, 1.235]	[0.560, 1.281]
β_1	[0.0905, 0.1195]	[0.0884, 0.1216]

The exact joint region is a tilted ellipse, since $\text{corr}(b_0, b_1) = -0.91$.

2. Mean response: one band for the whole line

Estimate $E\{Y_h\}$ at several X_h at once,
 $\hat{Y}_h \pm (\text{mult}) s\{\hat{Y}_h\}$:

- **Working–Hotelling:**

$W^2 = 2F(1 - \alpha; 2, n - 2)$, so $W = 2.463$.

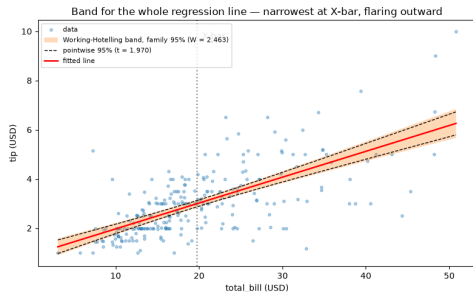
Holds for *all* X_h — the whole line — and does not depend on g .

- **Bonferroni:** $B = t(1 - \alpha/2g; n - 2)$, only for the g stated levels, and it grows with g .

- **Use the smaller.** The band is $W/t = 1.25 \times$ the pointwise interval ($t = 1.970$), at every X .

g	2	3	4	5	10	W
B	2.255	2.411	2.517	2.596	2.833	2.463

Bonferroni wins only for $g \leq 3$ here.



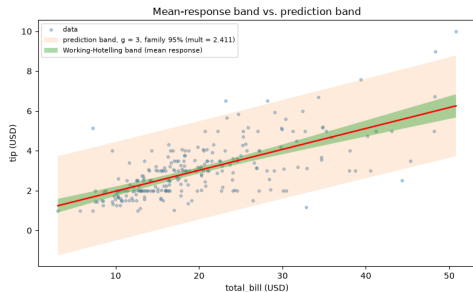
X_h	$\hat{Y}_h$	W-H, $g = 3$	Bonferroni, $g = 3$
10	1.971	[1.731, 2.210]	[1.736, 2.205]
20	3.021	[2.860, 3.182]	[2.863, 3.179]
30	4.071	[3.825, 4.317]	[3.831, 4.311]

3. Predicting new tips: the prediction band

For g new observations,

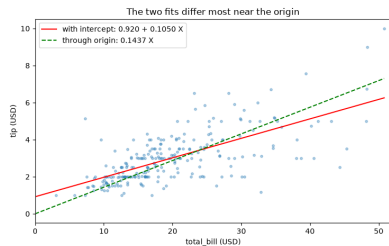
$s^2\{pred\} = MSE\left[\mathbf{1} + \frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2}\right]$: the extra 1 is the new tip's own scatter.

- **Scheffé** $S = 2.815$ vs. **Bonferroni** $B = 2.411$ ($g = 3$): use Bonferroni.
- At $\bar{X}$ the prediction band is about $15\times$ wider than the Working–Hotelling band.
- **Intervals can leave the possible range:** at $X_h = 10$ the lower limit is -0.51 , but a tip cannot be negative.
- **Constant width, growing scatter:** the pointwise 95% interval holds 97.5% of tips in the lowest third of bills but only 81.9% in the highest (93.0% overall).



X_h	$\hat{Y}_h$	$s\{pred\}$	family 95% PI
10	1.971	1.027	$[-0.505, 4.446]$
20	3.021	1.024	$[0.552, 5.490]$
30	4.071	1.027	$[1.595, 6.547]$

4. Other topics and takeaway



	Intercept	Origin
b_1	0.105	0.144
$s\{b_1\}$	0.0074	0.0032
MSE (df)	1.045 (242)	1.183 (243)
" R^2 "	0.457	0.892*

* uncentred; not comparable with 0.457.

- **Through the origin** is defensible on theory (zero bill, zero tip), but the data object: $t^* = 5.76$ for $\beta_0 = 0$.
- **Inverse prediction:** a \$5 tip gives $\hat{X} = 38.85$, interval $[19.5, 58.2]$ — wider than the observed bills (3–51); with $r = 0.676$ a tip pins the bill down poorly.
- **Design:** $s\{b_1\}$ would be 0.0316 with only the middle half of the bills (observed: 0.0074); $s\{\hat{Y}_h\}$ at $X_h = 45$ is $3\times$ its value at $\bar{X}$.

Takeaway. Several statements at once cost a wider interval: choose the procedure by g and by what is being claimed. Intervals for *new* observations also lean on constant variance and normality, which the earlier diagnostics put in doubt.